AP Precalculus **Unit 1, Lesson 1.14**

Group Activity — Answer Key

Name: ______________________ Partner: ____________________ Period: ________

The Engineering Design Challenge

Engineers use function models to make decisions. In each scenario below, you will **construct a model** and **apply it** to answer design questions. Work through your assigned tier. If you finish early, try the next tier.

Tier R — Remediate

Scenario: A rectangular solar panel array has one side against a wall. The available border is 60 m. The area function has already been set up for you:

$$A(w) = -2w^2 + 60w, \quad 0 < w < 30$$

where w is the width in meters and A is the area in square meters.

1. Evaluate the model at each width. Show your substitution.

w (m)	Work	$A(w)$ (m^2)
5	$-2(25) + 300$	**250**
10	$-2(100) + 600$	**400**
15	$-2(225) + 900$	**450**
25	$-2(625) + 1500$	**250**

2. Compute the average rate of change of A on the interval $[5, 15]$.

   $\text{ARC} = \dfrac{A(15) - A(5)}{15 - 5} = \dfrac{\mathbf{200}}{\mathbf{10}} = \mathbf{20}$

   Units: **m^2 per m (square meters per meter of width)**

3. Based on your table in part (1), between which two widths does the maximum area appear to occur?

   Between $w =$ **10** m and $w =$ **25** m **(vertex at $w = 15$; maximum is 450 m^2)**.

Tier A — Apply

Scenario: A storage company charges based on distance from a central depot. Testing shows that the shipping cost C (in dollars) is inversely proportional to the square of the distance d (in km) from the depot, with proportionality constant $k = 2000$.

1. Write the rational function model $C(d)$.

   $$C(d) = \frac{2000}{d^2}$$

2. State the domain restriction and explain it in context.

   Domain: $d > 0$ because **distance cannot be zero or negative — the depot must exist at a positive distance from the delivery point.**

3. Complete the table and describe the trend.

d (km)	2	4	10	20
$C(d)$ (\$)	**500**	**125**	**20**	**5**

   Trend: As distance increases, cost **decreases (approaches 0)**.

4. Compute the average rate of change of C on $[2, 4]$. Include units.

   $$\text{ARC} = \frac{C(4) - C(2)}{4 - 2} = \frac{\mathbf{-375}}{\mathbf{2}} = -187.50$$

   Units: **dollars per km (\$/km)**

Teacher Note

ARC $= -187.50$ \$/km: the negative sign indicates cost decreases as distance increases. Students should note this is the average decrease, not a steady rate (the model is not linear).

Tier E — Extend

Scenario: An engineer collects three data points for a project's profit P (in thousands of dollars) versus number of units produced x (in hundreds):

x	1	2	3
$P(x)$	-1	5	5

1. Assume $P(x) = ax^2 + bx + c$. Substitute each data point to write a system of three equations in a, b, c.

   $(1, -1)$: $a + b + c = -1$

   $(2, 5)$: $4a + 2b + c = 5$

   $(3, 5)$: $9a + 3b + c = 5$

2. Solve the system. Show your elimination or substitution steps.

   Teacher Note

   Subtract eq1 from eq2: $3a + b = 6$. Subtract eq2 from eq3: $5a + b = 0$. Subtract: $2a = -6 \Rightarrow a = -3$. Then $b = 0 - 5(-3) = 15$. Then $c = -1 - (-3) - 15 = -13$.

   $a = -3$ $\quad b = 15$ $\quad c = -13$

3. Write the complete quadratic model.

   $P(x) = -3x^2 + 15x - 13$

4. State one assumption this model makes that may not hold in reality.

   The model assumes profit follows a perfect quadratic pattern; real-world profit may fluctuate irregularly or not be symmetric.

5. Predict the profit when $x = 4$ (400 units). Is this prediction inside or outside the range of the data?

   $P(4) = -3(16) + 15(4) - 13 = -48 + 60 - 13 = -1$ **thousand dollars**. This is **outside (extrapolation beyond)** the data range.